WORKED SOLUTION

STEP I 2009, Question 6

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s1-q6

The key idea

The limits $1/m$ and m are swapped by $x \mapsto 1/x$, so that substitution is the natural one to try: it maps the interval to itself and trades a power x^{-n} for a power u^{n-1}. Part (ii) then lets us turn the awkward pieces of part (iii) into copies of the friendly integral from part (i). In (iii)(b) the interval is no longer symmetric, but the same substitution sends $[1,2]$ to $[\frac{1}{2},1]$, and the two halves glue back together.

Solution

Part (i)

Dividing x^2 by $x+1$,

$$\frac{x^2}{x+1} = x - 1 + \frac{1}{x+1},$$

so, since $x+1 > 0$ on the interval (as $m > 0$),

$$\int_{1/m}^{m} \frac{x^2}{x+1}\,\mathrm{d}x = \left[\tfrac{1}{2}x^2 - x + \ln(x+1)\right]_{1/m}^{m}.$$

The logarithmic part gives

$$\ln(m+1) - \ln\left(\frac{1}{m}+1\right) = \ln\frac{m+1}{(m+1)/m} = \ln m.$$

The polynomial part gives

$$\begin{aligned}\tfrac{1}{2}\left(m^2 - \frac{1}{m^2}\right) - \left(m - \frac{1}{m}\right) &= \left(m - \frac{1}{m}\right)\left[\tfrac{1}{2}\left(m + \frac{1}{m}\right) - 1\right] \\ &= \frac{m^2-1}{m}\cdot\frac{m^2-2m+1}{2m} \\ &= \frac{(m-1)(m+1)(m-1)^2}{2m^2} = \frac{(m-1)^3(m+1)}{2m^2}.\end{aligned}$$

Adding the two parts gives the required result.

Part (ii)

Put $x = 1/u$, so $\mathrm{d}x = -\mathrm{d}u/u^2$. When $x = 1/m$, $u = m$; when $x = m$, $u = 1/m$. Also

$$x^n(x+1) = \frac{1}{u^n} \cdot \frac{1+u}{u} = \frac{u+1}{u^{n+1}}.$$

Hence

$$\begin{aligned}\int_{1/m}^{m} \frac{1}{x^n(x+1)}\,\mathrm{d}x &= \int_{m}^{1/m} \frac{u^{n+1}}{u+1} \cdot \left(-\frac{1}{u^2}\right)\mathrm{d}u \\ &= \int_{1/m}^{m} \frac{u^{n-1}}{u+1}\,\mathrm{d}u,\end{aligned}$$

where the minus sign was used to swap the limits back. This is the required result.

Part (iii)

(a) Split the integrand:

$$\frac{x^5+3}{x^3(x+1)} = \frac{x^2}{x+1} + \frac{3}{x^3(x+1)}.$$

The interval is $[\frac{1}{2}, 2]$, which is part (ii)'s interval with $m = 2$. Using part (ii) with $n = 3$,

$$\int_{1/2}^{2} \frac{3}{x^3(x+1)}\,\mathrm{d}x = 3\int_{1/2}^{2} \frac{u^2}{u+1}\,\mathrm{d}u,$$

which is three times the first piece. By part (i) with $m = 2$,

$$\int_{1/2}^{2} \frac{x^2}{x+1}\,\mathrm{d}x = \frac{1^3 \cdot 3}{2 \cdot 4} + \ln 2 = \frac{3}{8} + \ln 2.$$

So the integral is $4(\frac{3}{8} + \ln 2)$, that is,

$$\int_{1/2}^{2} \frac{x^5+3}{x^3(x+1)}\,\mathrm{d}x = \frac{3}{2} + 4\ln 2.$$

(b) Now

$$\frac{x^5+x^3+1}{x^3(x+1)} = \frac{x^2}{x+1} + \frac{1}{x+1} + \frac{1}{x^3(x+1)}.$$

Ask yourself The interval is now $[1,2]$, not of the form $[1/m, m]$. Can part (ii) still help?

Yes: the substitution $x = 1/u$ works on any interval of positive numbers; it just moves the limits. Exactly the same computation as in part (ii), with $x = 1$ giving $u = 1$ and $x = 2$ giving $u = \frac{1}{2}$, shows

$$\int_{1}^{2} \frac{1}{x^3(x+1)}\,\mathrm{d}x = \int_{1/2}^{1} \frac{u^2}{u+1}\,\mathrm{d}u.$$

This has the same integrand as the first piece, on the neighbouring interval, so the two together make the integral of part (i) with $m = 2$:

$$\int_1^2 \frac{x^2}{x+1}\,\mathrm{d}x + \int_{1/2}^1 \frac{x^2}{x+1}\,\mathrm{d}x = \int_{1/2}^2 \frac{x^2}{x+1}\,\mathrm{d}x = \frac{3}{8} + \ln 2.$$

The remaining piece is

$$\int_1^2 \frac{1}{x+1}\,\mathrm{d}x = \ln 3 - \ln 2.$$

Adding,

$$\int_1^2 \frac{x^5+x^3+1}{x^3(x+1)}\,\mathrm{d}x = \frac{3}{8} + \ln 2 + \ln 3 - \ln 2 = \frac{3}{8} + \ln 3.$$

Checks

- In part (i), $m = 1$ gives an integral over a single point, and indeed the right-hand side is $0 + \ln 1 = 0$. Replacing m by $1/m$ should negate the integral (the limits swap), and the formula does change sign.
- Simpson's rule gives 4.272589 for (iii)(a), matching $\frac{3}{2} + 4\ln 2 \approx 4.272589$, and 1.473612 for (iii)(b), matching $\frac{3}{8} + \ln 3 \approx 1.473612$.
- Sanity check for (b): on $[1,2]$ the integrand dips from 1.5 to about 1.37 and then rises to about 1.71, so an answer near 1.47 is reasonable.

Common slips

- Forgetting the minus sign from $\mathrm{d}x = -\mathrm{d}u/u^2$, or forgetting to swap the limits; you must do both, and they cancel.
- In (iii)(b), applying part (ii) as if the interval were $[\frac{1}{2}, 2]$. The substitution moves $[1,2]$ to $[\frac{1}{2}, 1]$, and that is exactly what makes the pieces fit together.

Beyond STEP

The map $x \mapsto 1/x$ is a symmetry of the interval $[1/m, m]$, and using symmetries of the domain to relate integrals is a recurring trick in analysis. For example, it is how one shows $\int_0^\infty \frac{\ln x}{1+x^2}\,\mathrm{d}x = 0$. In complex analysis the same inversion $z \mapsto 1/z$ swaps the inside and outside of the unit circle and becomes a basic tool. Needham's *Visual Complex Analysis* has a beautiful geometric treatment of inversion.

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